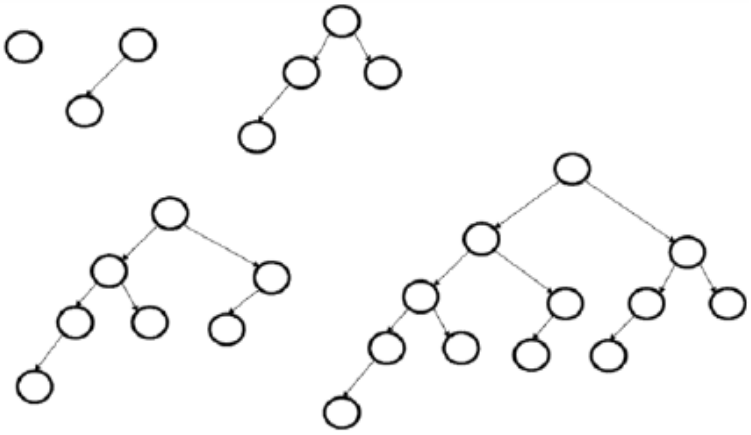
**Q1. (40pts)**

Draw an AVL tree topology of height *h=4* that contains the minimum number of nodes *N(4)*. How many nodes does this tree have?

Note that the height of a single-node AVL tree is *h=0* (i.e., *N(0)=1*)!

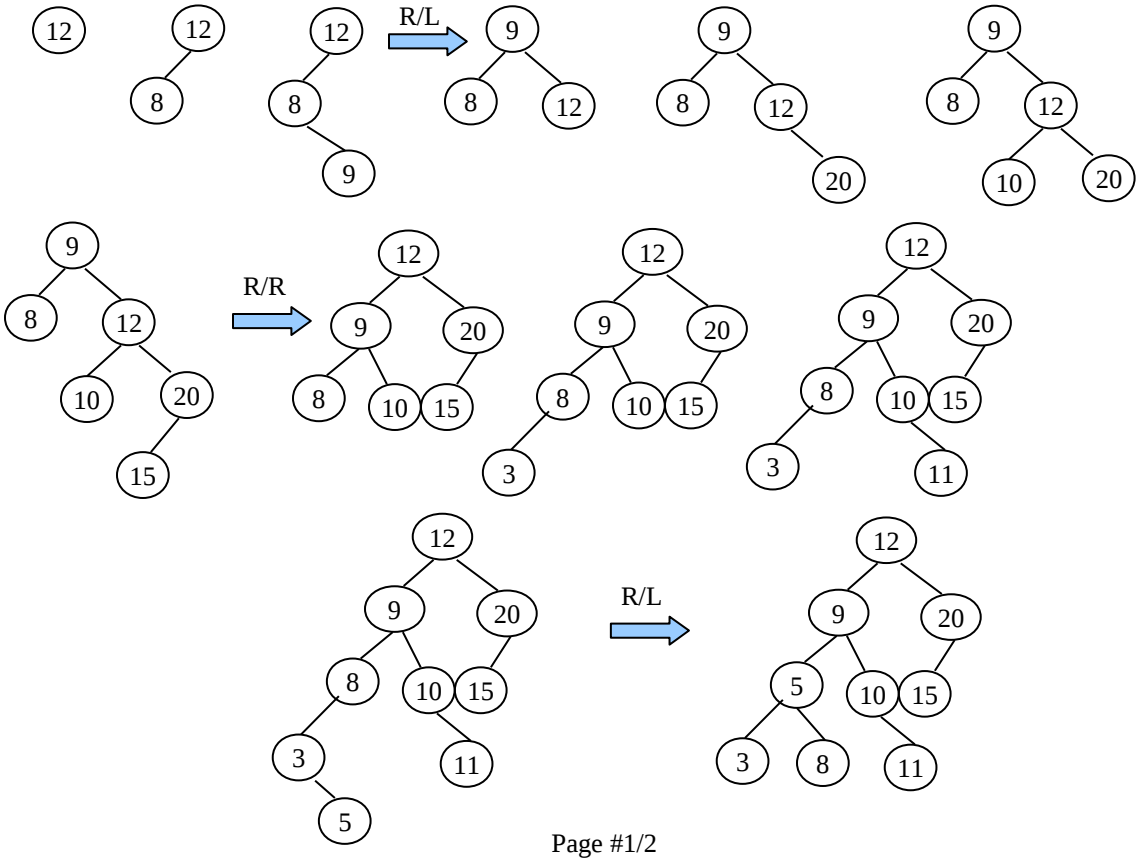
Construct a minimum size AVL tree of height h by creating a new root, and making one of its children a minimum AVL tree of height h-1, and the other a minimum AVL tree of h-2. So to get a minimum AVL tree of height 4, we need to build up minimum AVL trees of heights 0-3 first. The image below shows each of these, and finally a minimum AVL tree of height 4. Values are left out here, but any valid BST values could be filled in. Note that there are many different possible orderings of branches (left versus right); the ones shown below are one way to do it. The minimum number of nodes is 12.



**Q2. (40+20= 60pts)**

a) Insert the following keys, in order, into an initially empty AVL tree. Show each step!

12, 8, 9, 20, 10, 15, 3, 11, 5



b) Find a key we could insert into your resulting tree that would result in a L/L condition violation. Insert the key and restore the AVL property in the resulting tree.

Any key within the set {1,2,4,6,7,11,13,14} causes imbalance on the root.
Any key within the set {13,14} causes imbalance on the node with key 20.

